

Introduction and Potential Outcomes

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American University

2026-07-20

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Welcome

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- ▶ Associate Prof of Government
(American University)
- ▶ Associate Director, Center for Data Science
(American University)
- ▶ Senior Social Scientist
(The Lab @ DC)
- ▶ (Fellow in Methodology) (US Office of
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(Institut Barcelona d’Estudis Internacionals)
- ▶ Research agenda: political methodology,
causal inference, experimental design,
experiments in public policy

About You!

► Name?

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- ▶ Role?

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- ▶ Interests?

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- ▶ Role?
- ▶ Interests?
- ▶ A hobby or non-academic interest?

The Course: Learning Objectives

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- ▶ <https://github.com/ryantmoore/ci-obs-essex-2026/>

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- ▶ (Walk through syllabus)

What is Causal Inference?

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vs.

“What is the effect of foreign policy X on domestic terror attacks?”

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“Causes of an effect”

vs.

“Effects of a cause”

What is Causal Inference?

We will focus on “effects of a cause”

What is Causal Inference?

We will focus on “effects of a cause”, where the “cause” is well-defined.

Example: Canvassing and Programme Enrolment

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“Canvassing”: a systematic programme where agents talk with residents (knock on doors) at the residents’ home.



Figure 1: Credit: The Campaign Workshop

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- ▶ What fraction enrol under canvassing vs. no canvassing?
- ▶ $\frac{2}{2} - \frac{0}{2} = 1$
- ▶ (For each person canvassed, expect 1 more enrolment.)
- ▶ Did the policy “work” (cause more enrolment)?

Motivating Example: Canvassing and Enrolment

But, is it causal?

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What do we really want to know?

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Would canvassing actually *change* anyone's enrolment?

Motivating Example: Canvassing and Enrolment

But, is it causal?

What do we really want to know?

Would canvassing actually *change* anyone's enrolment?

What would have happened under *other* conditions?

Example: Canvassing and Enrolment

Citizen	Canvass?	Would Enrol if Canvass?	Would Enrol if No Canvass?	Enrol
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Empirical data consistent with *different unobserved* outcomes:

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What is the true causal effect of canvass?

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$$\frac{4}{4} - \frac{1}{4} = \frac{3}{4}$$

Example: Canvassing and Enrolment

Well ... how do we know which?

Example: Canvassing and Enrolment

Well ... how do we know which?

We can never know.

Can we know for one person?

Can we know for one person?

We can never know.

But I have some ideas.

But I have some ideas.

We could not canvass, then canvass later.

But I have some ideas.

We could not canvass, then canvass later.

We can never know.

We can never observe both “Canvassed”
and “Not Canvassed” for a unit.

We can never observe both “Canvassed”
and “Not Canvassed” for a unit.

We can never observe both
potential outcomes.

We can never observe both “Canvassed”
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We can never observe both the
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We can never know.

The Fundamental Problem of Causal Inference

We can never observe more than one
potential outcome for a given unit.

So, how can we get a *causal* estimate?

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We infer missing potential outcomes.

Correlation Is Not Causation

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Correlation is not enough for causality.

(For more, please enjoy

<https://www.tylervigen.com/spurious-correlations> and “spurious scholar”)

Causation Without Correlation

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- ▶ Wind blows; sailor turns rudder to *exactly* offset it

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Rudder does “nothing”!

But rudder *causes* straight path

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Because the sailor *endogenously* moves the rudder in response to the (unobserved) wind.

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Why visible under randomisation but not otherwise?

Because the sailor *endogenously* moves the rudder in response to the (unobserved) wind. The **randomness** helps reveal the (noisy) causal effect.

Optimisation Makes Behaviour Endogenous

Two more examples of optimising behaviour hiding a real effect:

- ▶ **Central bank** sees a recession forming and buys bonds to head it off. If done well, see *no* correlation between its actions and output — even though the policy worked.

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- ▶ **Central bank** sees a recession forming and buys bonds to head it off. If done well, see *no* correlation between its actions and output — even though the policy worked.
- ▶ **Student** who expects to gain most from course is one who enrolls. Comparing enrollees to non-enrollees confounds *effect* of course with *who chose* it.

How This Motivates the Course

- ▶ Experiments *randomise* — choice (Tr status) doesn't depend on potential outcomes

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Next, when our naïve estimate works

When Our Naïve Estimate Works

Why didn't we recover truth?

The problem with our naïve estimate of effect:

- ▶ “Canvass” group $\neq$ “No Canvass” group

Why didn't we recover truth?

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- ▶ The *potential outcomes* help predict whether Canvass!

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Knowing whether would enroll under canvass *predicts* whether canvassed!

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$$Pr(\text{Canvass} = \text{Yes} | \text{Would if Canvassed} = \text{Yes}) = \frac{2}{3}$$

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$$Pr(\text{Canvass} = \text{Yes} | \text{Would if Canvassed} = \text{Yes}) = \frac{2}{3}$$

$$\text{But } Pr(\text{Canvass} = \text{Yes} | \text{Would if Canvassed} = \text{No}) = 0$$

When comparing two groups **does** recover truth

Here, potential outcomes do **not** help predict Canvass:

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When comparing two groups **does** recover truth

Here, potential outcomes do **not** help predict Canvass:

Citizen	Canvass?	Would Enroll if Canvass?	Would Enroll if No Canvass?	Enroll
1	Yes	Yes	(Yes)	Yes
2	Yes	Yes	(No)	Yes
3	No	(Yes)	Yes	Yes
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Knowing whether enroll if canvass not predictive.

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Good! ☺

When does comparing groups recover truth?

Neither *potential outcome* should help predict treatment/intervention.

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True underlying responses in “Canvass” group =
True underlying responses in “No Canvass” group

(Note: *observed* outcomes can predict treatment.)

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- ▶ Aspirin $\implies$ headache!
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But, full schedule of true underlying responses should **not** predict treatment.

When does comparing groups recover truth?

How to ensure potential outcomes won't predict treatment?

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How to *assign* treatment so it won't predict potential outcomes?

When does comparing groups recover truth?

Possible *assignment mechanisms*:

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(We will estimate *correct* effect)

The Potential Outcomes Model

What is Causal Inference?

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*The objective is to determine for some population of units ...the ‘typical’ causal effect of the [treatment vs. control conditions] on a dependent variable Y .
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- ▶ A “causal effect” is a comparative statement

What is Causal Inference?

		Treatment Assignment	
		Randomized	Not Randomized
Unit Selection	Randomized	Randomized Experiment (gold standard)	Survey Sampling (allows population inference)
	Not Randomized	Controlled Experiment (allows causal inference)	Observational Study (large potential for bias)

Figure 2: Ramsey & Schafer, *The Statistical Sleuth*

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- ▶ One study, one causal effect (roughly)

The Potential Outcomes Model: Ideas

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- ▶ *Assignment mechanism*: means by which units come to be sorted into conditions

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Stable Unit Treatment Value Assumption (SUTVA)

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- ▶ No interference between units

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- ▶ For vector of Y_{i1} $\forall i$, write Y_1

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to represent the average of a *process* – the expectation.
Gerber and Green (2012) will use $E()$.

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If I roll 4 times and get 2, 6, 1, 1, the sample average is 2.5.

But the expected value of the process is still 3.5.

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Observed outcomes

► The observed outcome:

$$Y_i = Y_i(1) \cdot T_i + Y_i(0) \cdot (1 - T_i)$$

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The assignment mechanism *selects* which potential outcome we observe.

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Statistical language

- ▶ *parameter*: unknown numeric value characterising feature of prob model (Greek; θ , β)

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μ is my estimand. 5.1 is my estimate.

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For i , individual treatment effect

$$\tau_i = Y_i(1) - Y_i(0)$$

► True treatment effect?

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We can never know.

The Potential Outcomes Model: Estimands

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$$ATE = \overline{Y_1 - Y_0} = \overline{Y_1} - \overline{Y_0}$$

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$$ATE = \overline{Y_1} - \overline{Y_0} = \overline{Y_1} - \overline{Y_0}$$

- ▶ Average treatment effect for the treated

$$ATT = (\overline{Y_1}|T = 1) - (\overline{Y_0}|T = 1)$$

The Potential Outcomes Model: Estimands

The average treatment effect (ATE):

$$\begin{aligned}\bar{\tau} = \overline{TE} = ATE &\equiv \overline{Y_1 - Y_0} \\ &= \frac{1}{2n} \sum_{i=1}^{2n} (Y_{i1} - Y_{i0})\end{aligned}$$

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The Potential Outcomes Model: Estimands

Statistical Independence

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Define

$$Y \perp\!\!\!\perp X$$

as

$$Pr(Y|X) = Pr(Y)$$

The Potential Outcomes Model: Estimands

Statistical Independence

Define

$$Y \perp\!\!\!\perp X$$

as

$$Pr(Y|X) = Pr(Y)$$

(Knowing X doesn't change probability of Y)

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Something we **can** calculate:

If we know (Y_1, Y_0) indep of T

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$$\overline{Y_1} = (\overline{Y_1} | T = 1)$$

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The Potential Outcomes Model: Estimands

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Then, can substitute

$$\begin{aligned} ATE &= \overline{Y_1} - \overline{Y_0} \\ &= (\overline{Y_1} | T = 1) - (\overline{Y_0} | T = 0) \end{aligned}$$

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Then, can substitute

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Observed diff in Tr and Co group means gives $\overline{TE}$!

Why Experiments?

Random assignment of treatment promotes

$$(Y_1, Y_0) \perp\!\!\!\perp T$$

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$$(Y_1, Y_0) \perp\!\!\!\perp T$$

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$$(\overline{Y_1}|T = 1) \text{ and } (\overline{Y_0}|T = 0)$$

for things we can *never* know

$$\overline{Y_1} \text{ and } \overline{Y_0}$$

Common Assumptions, Null Hyp's in Causal Inference

► Constant effect:

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Common Assumptions, Null Hyp's in Causal Inference

- Constant effect:

$$\tau_i = Y_{i1} - Y_{i0} = \tau \quad \forall i$$

- Null hypothesis of no average effect:

$$ATE = \bar{\tau} = 0$$

- Sharp null hypothesis of no effect:

$$\tau_i = 0$$

Compliance with Treatment Assignment

Sometimes, units don't follow assignment!

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Assigned Tr	Assigned Co	Type
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Compliance with Treatment Assignment

Of course, we cannot observe i assigned both Tr and Co.

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		Treatment Taken	
		Treatment	Control
Tr Assigned	Tr	Complier/Always	Never/Defier
	Co	Always/Defier	Complier/Never

Common Estimates under Noncompliance

Let T_i be treatment *assigned*, D_i be treatment *received*.

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$$\begin{aligned}ITT &= (\overline{Y}_1|T=1) - (\overline{Y}_0|T=0) \\ &= (\overline{Y}_1|T=1, D(T=1)) - (\overline{Y}_0|T=0, D(T=0))\end{aligned}$$

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- ▶ Causal effect of *being assigned* (ignores receipt)

Common Estimates under Noncompliance

Let T_i be treatment *assigned*, D_i be treatment *received*.

- ▶ Intent-to-treat effect (as-assigned)

$$\begin{aligned}ITT &= (\overline{Y_1}|T=1) - (\overline{Y_0}|T=0) \\ &= (\overline{Y_1}|T=1, D(T=1)) - (\overline{Y_0}|T=0, D(T=0))\end{aligned}$$

- ▶ Causal effect of *being assigned* (ignores receipt)
- ▶ As-treated difference

$$ASTRE = (\overline{Y_1}|D=1) - (\overline{Y_0}|D=0)$$

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- ▶ Causal effect of *being assigned* (ignores receipt)
- ▶ As-treated difference

$$ASTRE = (\overline{Y_1}|D=1) - (\overline{Y_0}|D=0)$$

- ▶ If selection into D is confounded, likely *not* causal effect (ignores assignment)

Common Estimates under Noncompliance

Let T_i be treatment *assigned*, D_i be treatment *received*.

- ▶ Intent-to-treat effect (as-assigned)

$$\begin{aligned}ITT &= (\overline{Y_1}|T=1) - (\overline{Y_0}|T=0) \\ &= (\overline{Y_1}|T=1, D(T=1)) - (\overline{Y_0}|T=0, D(T=0))\end{aligned}$$

- ▶ Causal effect of *being assigned* (ignores receipt)
- ▶ As-treated difference

$$ASTRE = (\overline{Y_1}|D=1) - (\overline{Y_0}|D=0)$$

- ▶ If selection into D is confounded, likely *not* causal effect (ignores assignment)
- ▶ Per protocol difference

$$PP = (\overline{Y_1} | T=1, D=1) - (\overline{Y_0} | T=0, D=0)$$

Common Estimates under Noncompliance

Let T_i be treatment *assigned*, D_i be treatment *received*.

- ▶ Intent-to-treat effect (as-assigned)

$$\begin{aligned} ITT &= (\overline{Y_1} | T = 1) - (\overline{Y_0} | T = 0) \\ &= (\overline{Y_1} | T = 1, D(T = 1)) - (\overline{Y_0} | T = 0, D(T = 0)) \end{aligned}$$

- ▶ Causal effect of *being assigned* (ignores receipt)
- ▶ As-treated difference

$$ASTRE = (\overline{Y_1} | D = 1) - (\overline{Y_0} | D = 0)$$

- ▶ If selection into D is confounded, likely *not* causal effect (ignores assignment)
- ▶ Per protocol difference

$$PP = (\overline{Y_1} | T = 1, D = 1) - (\overline{Y_0} | T = 0, D = 0)$$

- ▶ If noncompliance, still confounded

Computing

Computing

- ▶ R for simulations, estimation, graphics

Computing

- ▶ R for simulations, estimation, graphics
- ▶ RStudio, an IDE for R

Computing

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- ▶ GitHub to store materials, submit PS

Computing

- ▶ R for simulations, estimation, graphics
- ▶ RStudio, an IDE for R
- ▶ GitHub to store materials, submit PS
- ▶ How should we chat? (email, Slack, ...)

R Primer

Functions

```
function(arg1, arg2, ...){  
  <the function code here...>  
}
```

Functions

```
function(arg1, arg2, ...){  
  <the function code here...>  
}
```

```
sum(5, 2)
```

```
[1] 7
```

Functions

```
function(arg1, arg2, ...){  
  <the function code here...>  
}
```

```
sum(5, 2)
```

```
[1] 7
```

```
mean(1:4)
```

```
[1] 2.5
```

Functions

```
nchar("greetings")
```


Functions

```
nchar("greetings")
```

```
[1] 9
```

Functions

```
nchar("greetings")
```

```
[1] 9
```

```
ls()
```

Functions

```
nchar("greetings")
```

```
[1] 9
```

```
ls()
```

```
character(0)
```

A Useful Function: `c()`

To concatenate objects into a vector, use `c()`:

A Useful Function: `c()`

To concatenate objects into a vector, use `c()`:

```
c(1, 3, 8, 20)
```

```
[1] 1 3 8 20
```

A Useful Function: `c()`

To concatenate objects into a vector, use `c()`:

```
c(1, 3, 8, 20)
```

```
[1] 1 3 8 20
```

```
c("essex", "summer", "school")
```

A Useful Function: `c()`

To concatenate objects into a vector, use `c()`:

```
c(1, 3, 8, 20)
```

```
[1] 1 3 8 20
```

```
c("essex", "summer", "school")
```

```
[1] "essex" "summer" "school"
```

A Useful Function: `c()`

To concatenate objects into a vector, use `c()`:

```
c(1, 3, 8, 20)
```

```
[1] 1 3 8 20
```

```
c("essex", "summer", "school")
```

```
[1] "essex" "summer" "school"
```

```
c(1, 2, "hello")
```


A Useful Function: `c()`

To concatenate objects into a vector, use `c()`:

```
c(1, 3, 8, 20)
```

```
[1] 1 3 8 20
```

```
c("essex", "summer", "school")
```

```
[1] "essex" "summer" "school"
```

```
c(1, 2, "hello")
```

```
[1] "1" "2" "hello"
```

Functions' Arguments

What arguments does a function have?

Functions' Arguments

What arguments does a function have?

```
help(median)  
args(median)
```

Functions' Arguments

What arguments does a function have?

```
help(median)  
args(median)
```

```
function (x, na.rm = FALSE, ...)  
NULL
```

Functions' Arguments

```
median(1:3)
```

```
[1] 2
```

Functions' Arguments

```
median(1:3)
```

```
[1] 2
```

```
x <- c(1, 2, 3, NA)  
median(x)
```

Functions' Arguments

```
median(1:3)
```

```
[1] 2
```

```
x <- c(1, 2, 3, NA)  
median(x)
```

```
[1] NA
```

Functions' Arguments

```
median(1:3)
```

```
[1] 2
```

```
x <- c(1, 2, 3, NA)  
median(x)
```

```
[1] NA
```

```
median(x, na.rm = TRUE)
```


Functions' Arguments

```
median(1:3)
```

```
[1] 2
```

```
x <- c(1, 2, 3, NA)  
median(x)
```

```
[1] NA
```

```
median(x, na.rm = TRUE)
```

```
[1] 2
```

Functions' Arguments

You can specify arguments in order or by name:

Functions' Arguments

You can specify arguments in order or by name:

```
median(x, TRUE)
```

```
[1] 2
```

Functions' Arguments

You can specify arguments in order or by name:

```
median(x, TRUE)
```

```
[1] 2
```

```
median(na.rm = TRUE, x)
```

```
[1] 2
```

Functions' Arguments

You can specify arguments in order or by name:

```
median(x, TRUE)
```

```
[1] 2
```

```
median(na.rm = TRUE, x)
```

```
[1] 2
```

```
median(TRUE, x)
```

```
Error in `if (na.rm) ...`:  
! the condition has length > 1
```

Some Useful Functions

Managing the workspace:

```
# Get the working directory ("Where am I?"):
getwd()
```

```
[1] "/Users/ryanmoore/Documents/github/ci-exp-  
essex-2026/notes"
```

Some Useful Functions

Managing the workspace:

```
# Get the working directory ("Where am I?):  
getwd()
```

```
[1] "/Users/ryanmoore/Documents/github/ci-exp-  
essex-2026/notes"
```

```
# Set the working directory:  
setwd("~/Desktop/")
```

Some Useful Functions

Managing the workspace:

```
# Get the working directory ("Where am I?):  
getwd()
```

```
[1] "/Users/ryanmoore/Documents/github/ci-exp-  
essex-2026/notes"
```

```
# Set the working directory:  
setwd("~/Desktop/")
```

But there are better ways: `.Rproj` and `{here}` package

Some Useful Functions

Managing the workspace:

```
# List objects in working dir:  
ls()
```

```
[1] "x"
```

```
# Remove `x' from working dir:  
rm(x)  
# Remove everything from working dir:  
rm(list = ls())
```

Some Useful Functions: Making vectors

```
c(1, 2, 10)
```

```
[1]  1  2 10
```

Some Useful Functions: Making vectors

```
c(1, 2, 10)
```

```
[1] 1 2 10
```

```
1:4
```

```
[1] 1 2 3 4
```

```
6:3
```

```
[1] 6 5 4 3
```

Some Useful Functions: Making vectors

```
seq(from = 5, to = 30, by = 5)
```

```
[1]  5 10 15 20 25 30
```

Some Useful Functions: Making vectors

```
seq(from = 5, to = 30, by = 5)
```

```
[1]  5 10 15 20 25 30
```

```
rep(c("a", "b"), 2)
```

```
[1] "a" "b" "a" "b"
```

Some Useful Functions: Making vectors

```
seq(from = 5, to = 30, by = 5)
```

```
[1]  5 10 15 20 25 30
```

```
rep(c("a", "b"), 2)
```

```
[1] "a" "b" "a" "b"
```

```
rep(c("a", "b"), each = 2)
```

```
[1] "a" "a" "b" "b"
```

Extracting Elements from Vectors with [

```
x <- sample(0:1, size = 10, replace = TRUE)
```

Extracting Elements from Vectors with [

```
x <- sample(0:1, size = 10, replace = TRUE)
```

```
x
```

```
[1] 1 1 0 1 0 0 1 0 1 1
```


Extracting Elements from Vectors with [

```
x <- sample(0:1, size = 10, replace = TRUE)
```

```
x
```

```
[1] 1 1 0 1 0 0 1 0 1 1
```

```
x[3]
```

Extracting Elements from Vectors with [

```
x <- sample(0:1, size = 10, replace = TRUE)
```

```
x
```

```
[1] 1 1 0 1 0 0 1 0 1 1
```

```
x[3]
```

```
[1] 0
```

Extracting Elements from Vectors with [

```
x <- sample(0:1, size = 10, replace = TRUE)
```

```
x
```

```
[1] 1 1 0 1 0 0 1 0 1 1
```

```
x[3]
```

```
[1] 0
```

```
x[3] <- 999
```

Extracting Elements from Vectors with [

```
x <- sample(0:1, size = 10, replace = TRUE)
```

```
x
```

```
[1] 1 1 0 1 0 0 1 0 1 1
```

```
x[3]
```

```
[1] 0
```

```
x[3] <- 999
```

```
x
```

```
[1] 1 1 999 1 0 0 1 0 1 1
```

Some Useful Functions: Making lists

```
v1 <- 1:8  
v2 <- letters[1:5]  
m <- matrix(1:9, 3, 3)
```

Some Useful Functions: Making lists

```
v1 <- 1:8  
v2 <- letters[1:5]  
m <- matrix(1:9, 3, 3)
```

v1

```
[1] 1 2 3 4 5 6 7 8
```

v2

```
[1] "a" "b" "c" "d" "e"
```

m

	[,1]	[,2]	[,3]
[1,]	1	4	7
[2,]	2	5	8
[3,]	3	6	9

Some Useful Functions: Making lists

```
l11 <- list(x = v1, y = v2, z = m)
```

Some Useful Functions: Making lists

```
l11 <- list(x = v1, y = v2, z = m)
```

Extract by name:

```
l11$y
```


Some Useful Functions: Making lists

```
l11 <- list(x = v1, y = v2, z = m)
```

Extract by name:

```
l11$y
```

```
[1] "a" "b" "c" "d" "e"
```

Some Useful Functions: Making lists

```
l11 <- list(x = v1, y = v2, z = m)
```

Extract by name:

```
l11$y
```

```
[1] "a" "b" "c" "d" "e"
```

Extract by position with `[[`:

```
l11[[1]]
```

Some Useful Functions: Making lists

```
l11 <- list(x = v1, y = v2, z = m)
```

Extract by name:

```
l11$y
```

```
[1] "a" "b" "c" "d" "e"
```

Extract by position with `[[`:

```
l11[[1]]
```

```
[1] 1 2 3 4 5 6 7 8
```

Some Useful Functions: Making lists

```
l1l <- list(x = v1, y = v2, z = m)
```

Extract by name:

```
l1l$y
```

```
[1] "a" "b" "c" "d" "e"
```

Extract by position with `[[`:

```
l1l[[1]]
```

```
[1] 1 2 3 4 5 6 7 8
```

Extract from sub-object:

```
l1l[[3]][2, 2]
```

Some Useful Functions: Making lists

```
l11 <- list(x = v1, y = v2, z = m)
```

Extract by name:

```
l11$y
```

```
[1] "a" "b" "c" "d" "e"
```

Extract by position with `[[`:

```
l11[[1]]
```

```
[1] 1 2 3 4 5 6 7 8
```

Extract from sub-object:

```
l11[[3]][2, 2]
```

```
[1] 5
```

Some Useful Mathematical Functions

```
5 + 2
```

```
[1] 7
```

```
5 - 2
```

```
[1] 3
```

```
5 * 2
```

```
[1] 10
```

```
5 / 2
```

```
[1] 2.5
```

Some Useful Mathematical Functions

```
5 ^ 2
```

```
[1] 25
```

```
sqrt(25)
```

```
[1] 5
```

```
20 %% 3
```

```
[1] 2
```

Some Useful Mathematical Functions and Values

```
pi
```

```
[1] 3.141593
```

```
abs(-3)
```

```
[1] 3
```

```
exp(1)
```

```
[1] 2.718282
```

```
log(exp(2))
```

```
[1] 2
```

```
sin(pi / 2)
```

```
[1] 1
```


Logicals

TRUE

```
[1] TRUE
```

FALSE

```
[1] FALSE
```

Logicals

```
TRUE
```

```
[1] TRUE
```

```
FALSE
```

```
[1] FALSE
```

```
TRUE == FALSE
```

Logicals

```
TRUE
```

```
[1] TRUE
```

```
FALSE
```

```
[1] FALSE
```

```
TRUE == FALSE
```

```
[1] FALSE
```

Logicals

```
c(1, 2) == c(1, 3)
```

Logicals

```
c(1, 2) == c(1, 3)
```

```
[1] TRUE FALSE
```

Logicals

```
c(1, 2) == c(1, 3)
```

```
[1] TRUE FALSE
```

```
c(1, 2) != c(1, 3)
```

Logicals

```
c(1, 2) == c(1, 3)
```

```
[1] TRUE FALSE
```

```
c(1, 2) != c(1, 3)
```

```
[1] FALSE TRUE
```

Logicals

```
c(1, 2) == c(1, 3)
```

```
[1] TRUE FALSE
```

```
c(1, 2) != c(1, 3)
```

```
[1] FALSE TRUE
```

```
c(1, 2) < c(1, 3)
```


Logicals

```
c(1, 2) == c(1, 3)
```

```
[1] TRUE FALSE
```

```
c(1, 2) != c(1, 3)
```

```
[1] FALSE TRUE
```

```
c(1, 2) < c(1, 3)
```

```
[1] FALSE TRUE
```

Logicals

```
c(1, 2) > c(1, 3)
```

```
[1] FALSE FALSE
```

```
c(1, 2) <= c(1, 3)
```

```
[1] TRUE TRUE
```

```
c(1, 2) >= c(1, 3)
```

```
[1] TRUE FALSE
```

How to Write a New Function

```
sumDiff <- function(num1 = 3, num2 = 5){  
  
  sum <- num1 + num2  
  
  diff <- num1 - num2  
  
  return(c(sum, diff))  
}
```

How to Write a New Function

```
sumDiff <- function(num1 = 3, num2 = 5){  
  
  sum <- num1 + num2  
  
  diff <- num1 - num2  
  
  return(c(sum, diff))  
}
```

Now, cut and paste function into R prompt.

How to Write a New Function

```
sumDiff <- function(num1 = 3, num2 = 5){  
  
  sum <- num1 + num2  
  
  diff <- num1 - num2  
  
  return(c(sum, diff))  
}
```

Now, cut and paste function into R prompt.
(R will tell you if syntax error.)

My New Function

```
sumDiff()
```

My New Function

```
sumDiff()
```

```
[1]  8 -2
```

My New Function

```
sumDiff()
```

```
[1]  8 -2
```

```
sumDiff(3, 5)
```


My New Function

```
sumDiff()
```

```
[1]  8 -2
```

```
sumDiff(3, 5)
```

```
[1]  8 -2
```

My New Function

```
sumDiff()
```

```
[1]  8 -2
```

```
sumDiff(3, 5)
```

```
[1]  8 -2
```

```
sumDiff(num2 = 5, num1 = 3)
```

My New Function

```
sumDiff()
```

```
[1]  8 -2
```

```
sumDiff(3, 5)
```

```
[1]  8 -2
```

```
sumDiff(num2 = 5, num1 = 3)
```

```
[1]  8 -2
```

My New Function

```
sumDiff()
```

```
[1]  8 -2
```

```
sumDiff(3, 5)
```

```
[1]  8 -2
```

```
sumDiff(num2 = 5, num1 = 3)
```

```
[1]  8 -2
```

```
sumDiff(5, 3)
```

My New Function

```
sumDiff()
```

```
[1]  8 -2
```

```
sumDiff(3, 5)
```

```
[1]  8 -2
```

```
sumDiff(num2 = 5, num1 = 3)
```

```
[1]  8 -2
```

```
sumDiff(5, 3)
```

```
[1]  8  2
```

My New Function

```
sumDiff(2, 20)
```

My New Function

```
sumDiff(2, 20)
```

```
[1] 22 -18
```

My New Function

```
sumDiff(2, 20)
```

```
[1] 22 -18
```

```
sumDiff(1, "yes")
```


My New Function

```
sumDiff(2, 20)
```

```
[1] 22 -18
```

```
sumDiff(1, "yes")
```

```
Error in `num1 + num2`:
```

```
! non-numeric argument to binary operator
```

Next:
Randomized Experiments,
Estimation, and Inference

Next:

Randomized Experiments,
Estimation, and Inference

Reading and PS1 for tomorrow!

Upload a `.R` or `.qmd` or `.Rmd` or `.txt` file to
<https://t.ly/1WuEn>
(must be logged in)

Gerber, Alan S., and Donald P. Green. 2012. *Field Experiments: Design, Analysis, and Interpretation*. New York, NY: WW Norton.